

Fractal Information Theory Project

FIN ST248–ST262

Preparation-Source Obstructions,
an Affine Carrier Connection, and Log-Haar Refinement

Analytical, interval, exact, and computational research report

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Abstract

Programs ST248–ST262 test whether the state, connection, refinement, and scale-flow objects identified in Release 10.73 can be generated or reduced further. ST248 proves a sharp obstruction: every deterministic D_{12} -natural probability state constructed from the strict operator A alone is uniform. Therefore the localized state used in $M_\rho = \text{diag } \rho$ cannot be hidden inside an equivariant function of A . ST249 then classifies all linear D_{12} -equivariant vertex-state processors as the seven-dimensional distance algebra and proves that they can preserve or erase, but cannot create, selector information absent from their input state.

Several positive closures follow. ST250 certifies 360 additional pseudo-arclength boxes, giving eight local steps on every signed branch. ST251 proves the exact global recovery optimum 81/100 for a rational, fully non-diagonal, nonunital rotated amplitude-damping channel. ST253 proves that A induces a positive metric and a canonical flat Levi-Civita connection on the mean-zero affine amplitude carrier; this connection cancels the nonlinear chart artifact found in ST239. ST254 certifies every one of $28^3 = 21,952$ nuisance boxes at halfwidth 0.0005. ST255 constructs an exactly finite-sample efficient modal estimator. ST257 gives a direct outward belief-transfer spectral-radius enclosure and improves the certified Hellinger-rate interval after intersection with the independent previous bound.

At the refinement frontier, graph-Laplacian locality reduces the abstract 78-dimensional two-child moduli space to a seven-dimensional positive cone but does not select one member. Complement-resolving preparations and effects reconstruct the hidden generator exactly, whereas every coarse experiment is blind to it. Finally, ST260 proves that the unique scale-invariant rate measure is the log-Haar measure $\rho d\mu/\mu$ and that it contributes the constant $2\rho \ln 2$ to spectral dimension. This gives a precise mathematical form to a fractal-layer plateau, but its density, cutoffs, realization, and physical units remain additional data. No selector closure, dimensional physics, laboratory evidence, legacy-to-strict role transfer, Standard Model, gravity, or Theory-of-Everything closure is claimed.

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1 Executive conclusions

The research batch yields four structural conclusions.

1. **The preparation obstruction is exact.** If the only input is the fully D_{12} -symmetric strict operator, deterministic naturality forces the output probability state to be uniform. A localized state requires a boundary, an orbit label, a random realization, or a genuinely new symmetry-breaking object.
2. **A nonlinear response connection exists conditionally.** On the canonical mean-zero affine amplitude carrier, A is a positive metric and its Levi-Civita connection is flat and natural. This solves nonlinear chart transport mathematically inside that carrier, but it is not a spacetime or gauge connection.
3. **Locality reduces refinement freedom without removing it.** Exact coarse diffusion plus graph-Laplacian signs and the declared symmetries leave six independent edge splits and one vertical rate.
4. **A fractal plateau has a unique scale-invariant measure.** The positive multiplicative group has Haar measure $d\mu/\mu$, and its integrated fiber contribution is independent of time. What is missing is no longer the mathematical shape of the plateau, but a strict source for its density, cutoffs, and operational realization.

Thus the deepest surviving statement is

A determines symmetric dynamics and a quotient geometry,
 A does not determine a localized state or a unique refinement.

2 Scope, kernel discipline, and proof labels

The strict working object remains

$$A = sI - W, \quad W_{xx} = 0, \quad W_{xy} = K_{\text{strict}}(d(x, y)) \quad (x \neq y),$$

with $A\mathbf{1} = 0$ and $A \succeq 0$. The diagonal exclusion is part of the definition. The historical $K_{\text{legacy,ont}}$ remains an intermediate bridge kernel; it is not silently identified with K_{strict} , and no legacy physical-role formula is transferred.

The project ontology treats the nadsoliton itself as primordial fractal information in a solitonic state. The present report analyzes mathematical consequences of a finite strict operator. It does not infer that a physical universe, matter, or spacetime has been derived.

- **[PROVEN]** means an exact deduction or a finite outward certificate within its stated class.
- **[STRONG NUMERICAL EVIDENCE]** means a reproducible numerical result without an exact or interval global-optimality proof.
- **[CONDITIONAL]** means that the stated carrier, state, control cost, channel class, or scale measure is an explicit premise.
- **[BLOCKED]** means that the needed evidence or source object is absent.

3 Program ledger

Pro-gram	Object	Result	Boundary
ST248	Preparation source	[PROVEN] deterministic D_{12} -natural no-go	New symmetry-breaking classes remain
ST249	Typed second operators	[PROVEN] seven-dimensional linear classification	Nonlinear maps open
ST250	Branch extension	[PROVEN] 360 new local boxes	Explicit finite stop
ST251	Nonunital recovery	[PROVEN] global optimum 81/100	Rotated amplitude damping class
ST252	Selector ledger	[STRONG NUMERICAL EVIDENCE] feasible augmented cost 0.440546	No interval KKT certificate
ST253	Carrier connection	[CONDITIONAL] flat Levi-Civita completion	Affine amplitude carrier supplied
ST254	Nuisance cover	[PROVEN] 21,952/21,952 boxes	Not a maximal domain
ST255	Mode estimator	[PROVEN] exact finite-sample efficiency	Declared local multinomial model
ST256	Joint Clifford pair	[PROVEN] aligned two-gap uniqueness	Generic misalignment open
ST257	HMM transfer	[PROVEN] outward spectral-radius enclosure	Fixed $s = 1/2$ HMM
ST258	Thermal reset	[PROVEN] ideal pulse and Gibbs error law	Controller and physical units supplied
ST259	Local refinement	[PROVEN] seven-dimensional positive cone	No member selected
ST260	Scale plateau	[CONDITIONAL] exact log-Haar theorem	Density and cutoffs unsourced
ST261	Refinement observables	[PROVEN] coarse no-go and complement tomography	Fiber instrument not supplied
ST262	External evidence	[BLOCKED]	No independent laboratory record

4 ST248–ST249: where the selector information must enter

4.1 The strict preparation-source no-go

Let F be a deterministic rule assigning a probability vector to a self-adjoint vertex operator, and impose naturality

$$F(gAg^{-1}) = gF(A), \quad g \in D_{12}.$$

The strict A is fixed by every g . Therefore $F(A) = gF(A)$ for the complete transitive action. The only invariant probability vector is $\mathbf{1}/12$.

Theorem 4.1 (No localized state from symmetric A alone). *No deterministic D_{12} -natural state-valued function of A can generate the localized state required by the noncommuting multiplication operator M_ρ .*

The diagonals of $e^{-0.7A}$, $(I + A)^{-1}$, and A^2 , after normalization, replay the theorem numerically to errors below 3×10^{-16} . By contrast, $e^{-0.7A}\delta_0$ lies 0.323238 from uniform and gives $\|[A, M_\rho]\|_F = 0.265194$. The difference is exactly the supplied preparation δ_0 .

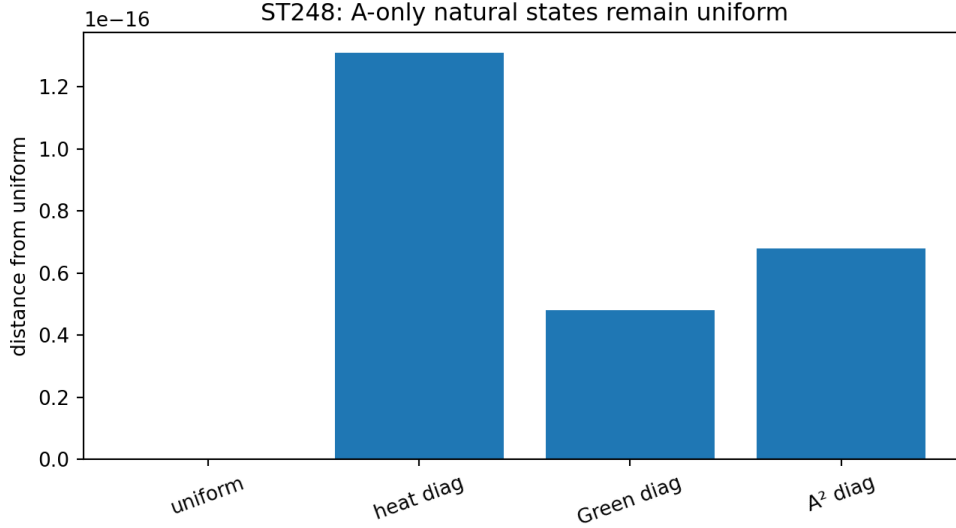


Figure 1: ST248: state candidates generated without an orbit label remain uniform.

This is a no-go for a declared class, not for spontaneous stochastic realization or a new strict symmetry-breaking field. QW-2191 remains open.

4.2 Complete linear typed-operator classification

The vertex permutation representation of D_{12} decomposes multiplicity-free into seven real distance sectors. Hence every linear equivariant state processor is

$$C = \sum_{d=0}^6 c_d B_d,$$

where B_d is the distance- d adjacency matrix. Positivity and mass preservation cut out the affine Markov polytope. Every typed multiplication operator in this class is

$$\rho \mapsto M_{C\rho} = \text{diag}(C\rho).$$

Equivariance implies

$$\text{Stab}(\rho) \subseteq \text{Stab}(C\rho).$$

Thus processing may erase asymmetry but cannot create orbit information absent from the condition. Uniform, even-localized, and chiral inputs yield stabilizer orders 24, 2, and 1, respectively. The chiral state is the minimal orbit type in this class capable of carrying both localization and orientation.

5 ST250: bounded continuation of every signed branch

Starting from the two-step endpoints of ST236, six further pseudo-arclength correctors were validated for each of sixty signed seeds. All 360/360 new boxes pass. Every complete seed now has eight certified steps. The minimum new rank margin remains positive, and the minimum clearance between boxes from different exact fold pairs is

$$0.0094481743.$$

No nonfold intersections occur in the traced extension, so the finite incidence graph still has thirty components.

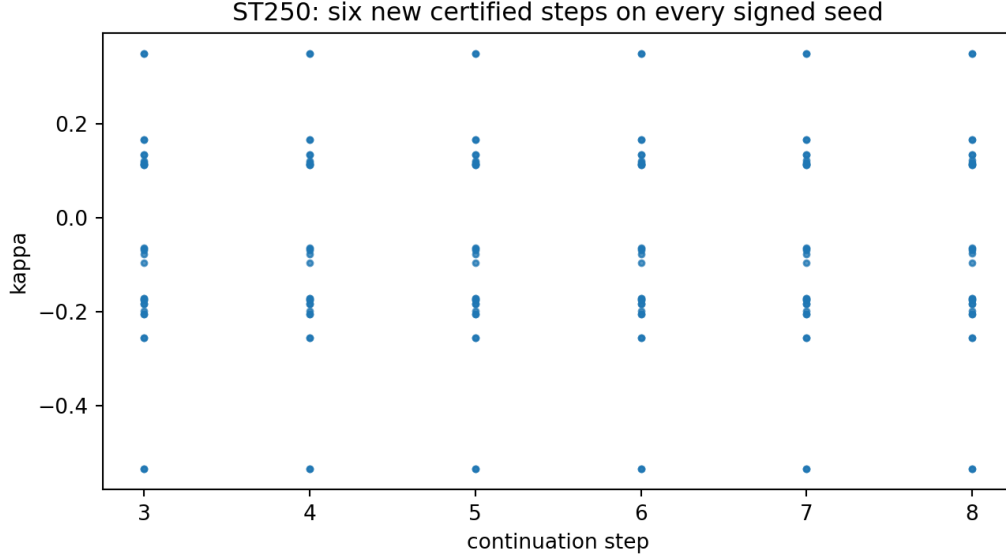


Figure 2: ST250: six new outward-certified steps on every signed seed.

The stop after six new steps is explicit. Nothing here excludes a later collision, secondary bifurcation, or distant connection, and the branches are not identified with physical particles.

6 ST251: a fully non-diagonal nonunital recovery theorem

Take amplitude damping with rational Kraus operators

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & 4/5 \end{pmatrix}, \quad K_1 = \begin{pmatrix} 0 & 3/5 \\ 0 & 0 \end{pmatrix},$$

and compose it with rational pre- and post-unitaries

$$U = \frac{1}{5} \begin{pmatrix} 3 & 4 \\ -4 & 3 \end{pmatrix}, \quad V = \frac{1}{13} \begin{pmatrix} 5 & 12 \\ -12 & 5 \end{pmatrix}.$$

The resulting affine Bloch map is nonunital and has off-diagonal linear norm 0.892378.

Theorem 6.1 (Global CPTP optimum). *The optimal recovery over every CPTP qubit map is*

$$R = U^* V^* = \frac{1}{65} \begin{pmatrix} -33 & -56 \\ 56 & -33 \end{pmatrix}, \quad F_e^* = \frac{81}{100}.$$

For the base damping channel, the exact dual is

$$Y = \text{diag}(9/20, 9/25).$$

Its trace is 81/100, and the slack consists of two positive scalar blocks and $\frac{1}{5} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$. Rational unitary transport gives the displayed channel primal and dual with zero exact gap. This extends the previous Pauli and replacer families, but not all affine channels.

7 ST252: endpoint and actuator costs

The ST238 two-state selector model was augmented by ten control knots, a hard dimensionless slew bound $|\dot{\pi}| \leq 0.30$, a quadratic actuator term, and endpoint KL costs relative to the baseline state. The ledger is

$$J = \Sigma_{\text{Markov}} + J_{\text{slew}} + J_{\text{endpoints}}.$$

Three independent starts converge to values agreeing within 3×10^{-13} . The best feasible schedule gives

$$\begin{aligned} J &= 0.4405461745, & \Sigma &= 0.3347718379, \\ J_{\text{slew}} &= 0.0054698773, & J_{\text{endpoints}} &= 0.1003044593. \end{aligned}$$

The final-state residual is 1.2×10^{-14} , and the slew constraint is active.

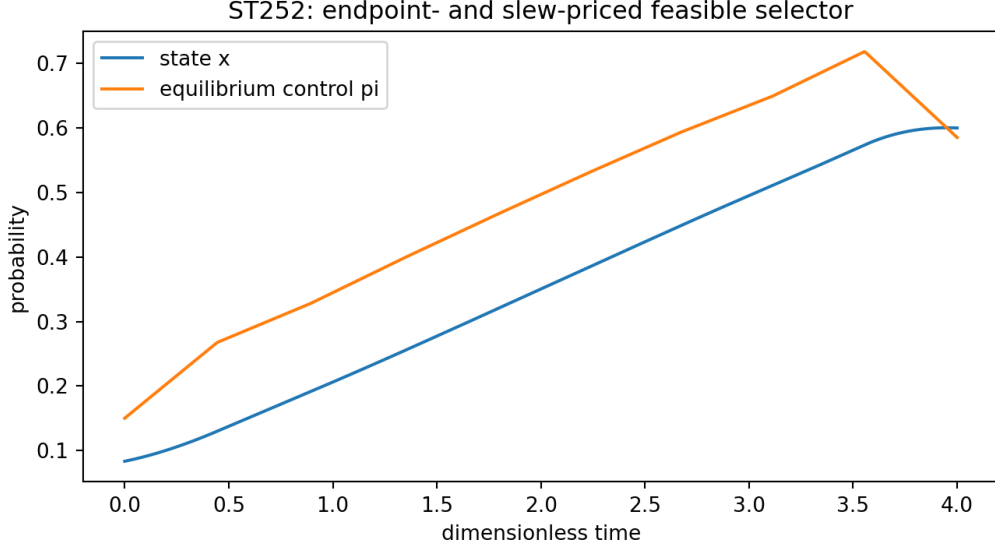


Figure 3: ST252: a feasible selector path after endpoint and finite-rate actuator pricing.

All added costs are nonnegative, so the verified ST238 lower bound 0.3176910887 remains a rigorous lower bound. The new schedule is only a numerical upper bound; without an interval KKT certificate, global optimality is not claimed. None of the coefficients are physical joules.

8 ST253: the mean-zero affine carrier connection

Because $\ker A = \text{span } \mathbf{1}$, the strict operator induces a positive metric on

$$H_0 = \{\psi \in \mathbb{R}^{12} : \mathbf{1}^T \psi = 0\}, \quad g_A(u, v) = u^T A v.$$

The metric eigenvalues lie between 0.754121 and 2.342182. On the affine vector space H_0 , the translation-invariant metric has a unique Levi-Civita connection. It is D_{12} -invariant, flat, and torsion-free.

The nonlinear scalar audit uses $x = y + cy^6$ and

$$V(x) = \frac{a}{2}x^2 + \frac{g}{12}x^{12}.$$

Ordinary differentiation gives

$$\frac{D_y^{12}V}{11!} = g + 6ac^2.$$

Transporting the flat connection gives $\Gamma_y = x''/x'$ and the recursion

$$T_{k+1} = \partial_y T_k - k\Gamma_y T_k.$$

Exact symbolic evaluation yields

$$\frac{\nabla_y^{12}V}{11!} = g.$$

The chart artifact cancels exactly.

This is a meaningful completion of ST239, conditional on the affine mean-zero amplitude carrier. It is not a physical spacetime, gravitational, or gauge connection.

9 ST254: complete nuisance-cover extension

The cube centered at $(0.2, 0.7, 0.05)$ with halfwidth 0.0005 was tiled by $28^3 = 21,952$ closed boxes. Every outward Krawczyk test passes. The minimum inclusion margin is

$$1.4721098531 \times 10^{-4},$$

and the maximum is $2.0794075222 \times 10^{-4}$. This converts the sampled ST240 diagnosis into a complete cube theorem.

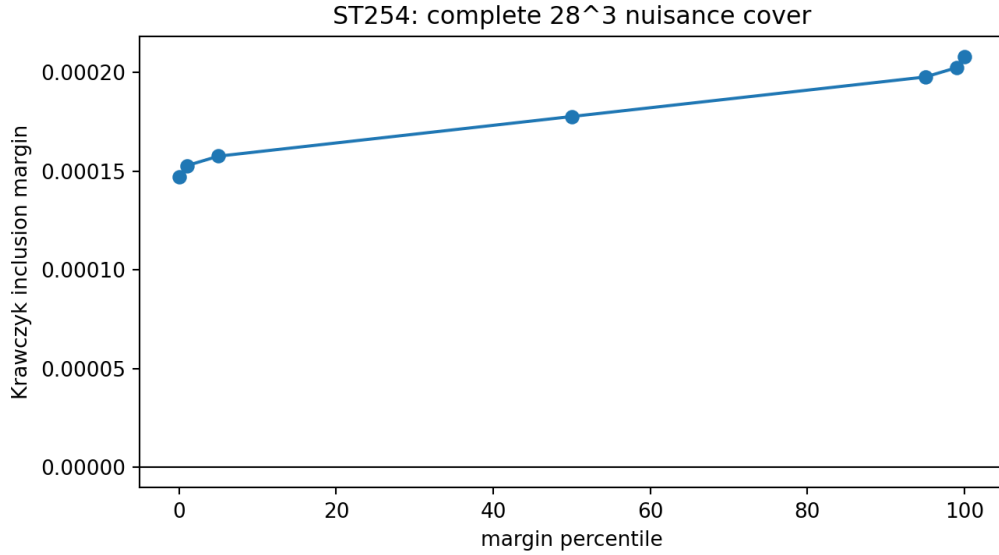


Figure 4: ST254: inclusion-margin percentiles over the complete 28^3 cover.

The result is a continuation domain, not an experimental tolerance or a proof that the domain is maximal.

10 ST255–ST257: exact estimation and transfer-operator inference

10.1 Finite-sample efficient modal estimators

For orthonormal mean-zero Fourier modes v_k and attenuations σ_k , consider the local multinomial family

$$p(\theta) = \frac{\mathbf{1}}{12} + \sum_{k=1}^6 \sigma_k \theta_k v_k.$$

The estimator

$$\hat{\theta}_k = \frac{v_k^T (\hat{p} - \mathbf{1}/12)}{\sigma_k}$$

is exactly unbiased wherever the probabilities are nonnegative. At the uniform state,

$$\text{Cov}(\hat{\theta}) = \text{diag} \left(\frac{1}{12n\sigma_k^2} \right) = \frac{I^{-1}}{n}.$$

It attains the finite-sample Cramér–Rao matrix exactly. In the supplied common-shock cluster model, the covariance is multiplied exactly by $1 + (20 - 1)0.05 = 1.95$.

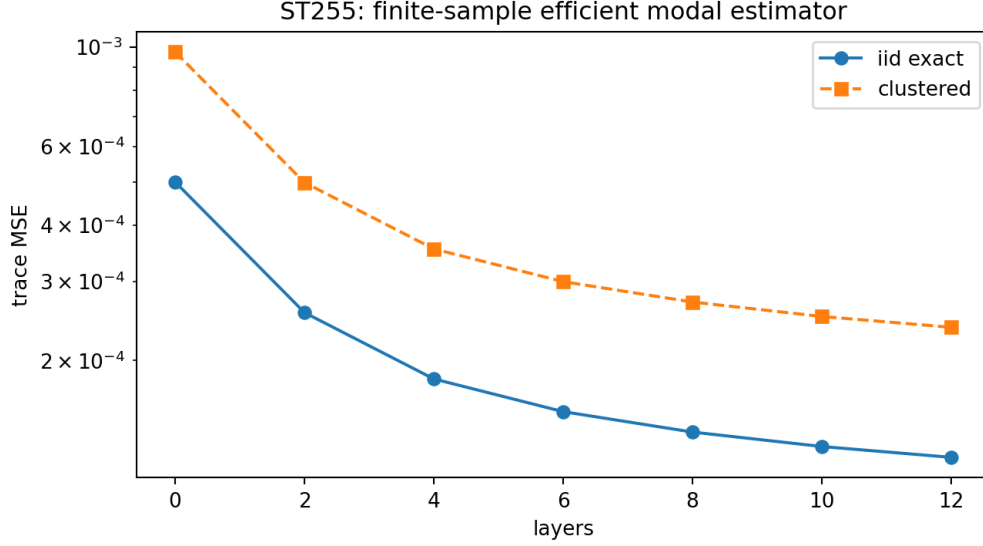


Figure 5: ST255: exact trace MSE and cluster inflation across developed layers.

10.2 An aligned two-gap Clifford theorem

Let (X_*, Z_*) be any balanced Hermitian Clifford pair and let

$$X_0 = \alpha X_*, \quad Z_0 = \beta Z_*, \quad \alpha, \beta > 0.$$

For every competing Clifford pair,

$$\Delta J = \alpha \|X - X_*\|_F^2 + \beta \|Z - Z_*\|_F^2.$$

Thus (X_*, Z_*) is the unique global joint minimizer in every balanced dimension. This is an exact global theorem for the aligned two-gap class and is invariant under common unitary conjugation. Marginal sign gaps alone for arbitrary misaligned targets are not settled.

10.3 Direct belief-transfer spectral radius

For the observed HMM, define the positive transfer operator

$$(Tf)(b_0, b_1) = \sum_y \sqrt{q_{0,y}(b_0)q_{1,y}(b_1)} f(F_{0,y}(b_0), F_{1,y}(b_1)).$$

Adjacent floating intervals enclose every exact decimal parameter. A 24^2 cover of the complete belief square and depth-nine outward recursion give

$$T^9 \mathbf{1} \in [0.00058633896, 0.00166319692].$$

For a positive operator, $m\mathbf{1} \leq T^n \mathbf{1} \leq M\mathbf{1}$ implies

$$m^{1/n} \leq r(T) \leq M^{1/n}.$$

Hence

$$r(T) \in [0.43742683, 0.49115206], \text{quad}\mathcal{C}_{1/2} \in [0.71100152, 0.82684584].$$

Intersecting with the independent previous certificate yields

$$\boxed{\mathcal{C}_{1/2} \in [0.7110015196, 0.8198470893]},$$

of width 0.10884557. This is a direct interval transfer result, not the strong-numerical Chernoff optimization of ST243.

11 ST258: ideal controller pulse and finite-temperature blanks

For two identical registers let S denote global SWAP and choose

$$H_c = \frac{\pi}{2\tau}(I - S), \quad 0 < t < \tau.$$

Then

$$e^{-i\tau H_c} = S, \text{quad}[H_c, H_{\text{total}}] = 0.$$

The expectation of H_c is constant during the pulse, so ideal sudden switching on and off has zero net closed-system work. This does not include dissipation in a finite controller.

A Gibbs blank with dimensionless gap $\beta\epsilon$ has residual reset error

$$\delta = \frac{1}{1 + e^{\beta\epsilon}}, \quad \beta\epsilon = \log \frac{1 - \delta}{\delta}.$$

Exact pure reset requires $\beta\epsilon \rightarrow \infty$. The blank is therefore a thermodynamic resource rather than a free logical symbol.

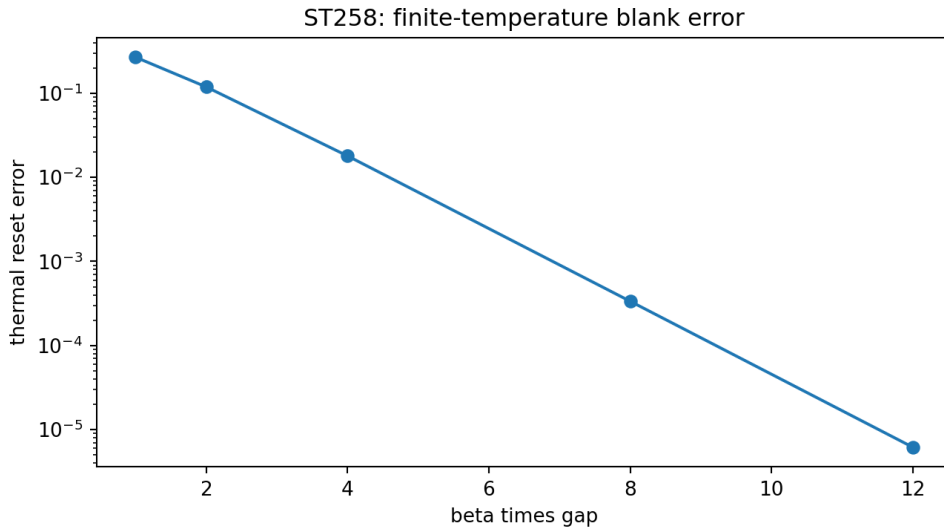


Figure 6: ST258: finite-temperature reset error as a function of the dimensionless gap.

12 ST259: the local positive refinement cone

Label refined vertices by (i, a) with base vertex $i \in \mathbb{Z}_{12}$ and fiber $a \in \{0, 1\}$. Let p_d be the same-fiber weight, q_d the cross-fiber weight at base distance d , and v the vertical weight. Translation, reflection, and fiber swap leave these parameters invariant.

Theorem 12.1 (Local graph-Laplacian classification). *The refined graph Laplacian exactly intertwines the coarse strict generator if and only if*

$$p_d \geq 0, \quad q_d \geq 0, \quad p_d + q_d = W_d \quad (d = 1, \dots, 6), \quad v \geq 0.$$

Thus the 78-dimensional abstract symmetric complement cone collapses to six closed intervals times one ray. All sampled matrices have nonpositive off-diagonals, zero row sum, positive spectrum, and intertwining error below 2.1×10^{-15} . If same- and cross-fiber edges are declared indistinguishable, $p_d = q_d = W_d/2$ and only v remains. Neither symmetry nor locality chooses the remaining parameters.

13 ST260: the log-Haar plateau theorem

Each two-state fiber gap μ contributes

$$f(\mu t) = \frac{4\mu t}{e^{2\mu t} + 1}$$

to spectral dimension. Exact invariance under scale multiplication selects, up to a constant density ρ , the Haar measure of $\mathbb{R}_{>0}$:

$$d\nu(\mu) = \rho \frac{d\mu}{\mu}.$$

Theorem 13.1 (Scale-invariant plateau). *For every $t > 0$,*

$$\int_0^\infty \frac{4\mu t}{e^{2\mu t} + 1} \rho \frac{d\mu}{\mu} = 2\rho \ln 2.$$

The proof substitutes $x = 2\mu t$ and uses $\int_0^\infty (e^x + 1)^{-1} dx = \ln 2$. A finite dyadic ladder $\mu_j = 2^j$, $-12 \leq j \leq 12$, has logarithmic density $1/\ln 2$ and therefore approaches the plateau value 2. Numerically it stays within 0.02 of 2 for

$$4.01 \times 10^{-4} \lesssim t \lesssim 41.04.$$

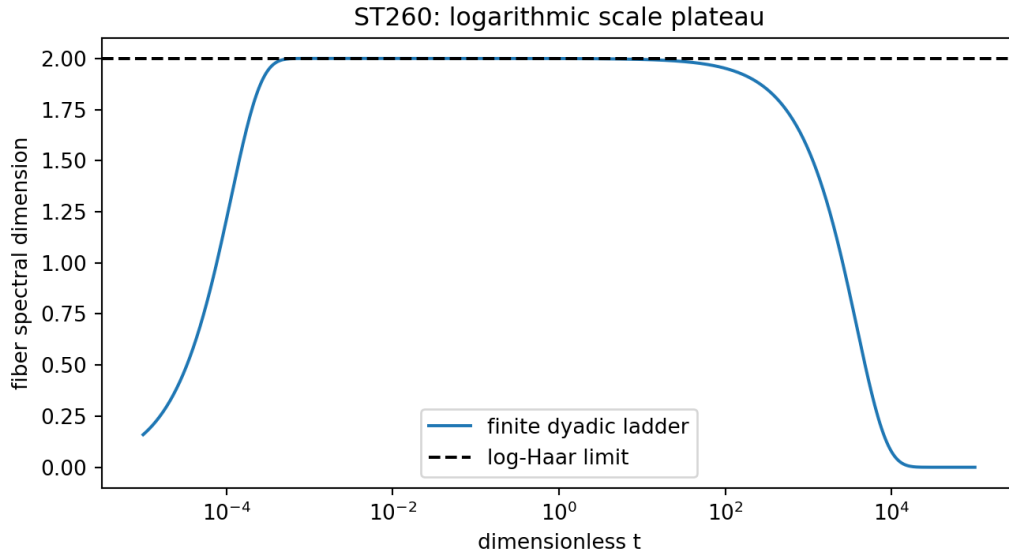


Figure 7: ST260: finite dyadic scale flow and its log-Haar continuum plateau.

This is the most direct mathematical realization so far of the intuition that observation may occur through many self-similar compression layers. It does not identify which layer we inhabit. The density, finite cutoffs, tensor/trace completion, and a physical mapping from dimensionless t to seconds or length remain missing. The dyadic illustration is not a license to transfer legacy octave or α_{geo} roles into the strict kernel.

14 ST261: which observations can see a deeper refinement?

Let R embed the coarse space and S embed its orthogonal fiber complement. For every exact self-adjoint refinement

$$\tilde{A} = RAR^* + SBS^*,$$

functional calculus gives

$$e^{-t\tilde{A}}R = Re^{-tA}, \quad S^*e^{-t\tilde{A}}S = e^{-tB}.$$

Theorem 14.1 (Coarse blindness and complement tomography). *Every experiment whose preparations lie in $\text{Ran } R$ and whose effects are coarse-compressed is exactly independent of B . Conversely, a complement-resolving preparation and measurement reconstruct e^{-tB} . At one known $t > 0$,*

$$B = -\frac{1}{t} \log(S^* e^{-t\tilde{A}} S)$$

is unique.

For complement rates $\mu = 0.3, 1, 4$, all coarse return probabilities agree to 10^{-15} , while the odd-fiber returns are $e^{-\mu}$ and are plainly separated.

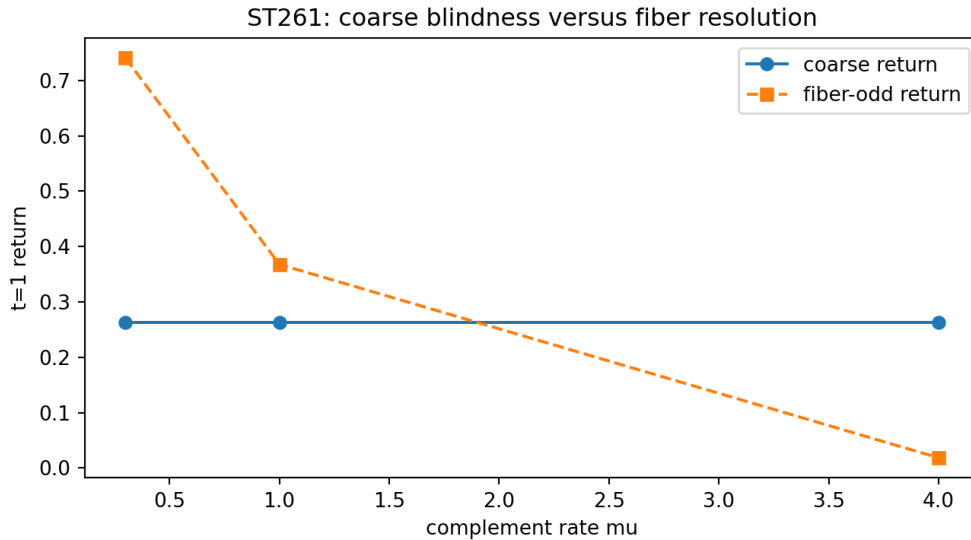


Figure 8: ST261: exact coarse blindness versus complement-sensitive survival.

The theorem identifies the missing operational object: a fiber-resolving preparation and instrument. It does not assert that such a physical fiber exists.

15 Falsification ledger

Candidate claim	Test result	Consequence
A naturally prepares a localized state	Transitive naturality forces uniform	Strict deterministic preparation source refuted
Equivariant filtering creates orientation	Stabilizer can only grow	Input condition carries asymmetry
The nonlinear response lacks any repair	Flat quotient connection cancels artifact	Repair exists conditionally on carrier
The halfwidth-0.0005 cube may lose roots	All 21,952 boxes pass	Prior coarse failures were methodological
Finite-sample efficiency is only asymptotic	Explicit modal estimator attains CR exactly	Exact result in declared multinomial model
Two marginal gaps solve every joint factor problem	Only aligned two-gap class proved	Generic misalignment remains open
Locality selects one refinement	Seven-dimensional positive cone remains	Nonuniqueness reduced, not removed
Coarse observations can infer hidden rates	Exact intertwining erases all B dependence	Fiber-resolving instruments are necessary

Self-similar rates can form no stable plateau	Log-Haar integral is constant	Plateau shape proved, source not proved
Local computation supplies physical evidence	No external record exists	ST262 remains blocked

16 Interpretation for FIN and layered observation

The combined ST248–ST261 picture is internally coherent. The strict generator determines a symmetric semigroup and a positive quotient geometry. A localized state creates a second noncommuting algebra, but cannot be obtained by a deterministic natural rule from A alone. Refinement introduces complement dynamics invisible to every coarse experiment. A fiber-resolving instrument can reveal them. A log-Haar distribution of fiber rates creates a broad scale-invariant plateau.

This supports a precise version of the “observation through fractal layers” hypothesis: coarse observers can be exactly blind to complement dynamics while a logarithmic hierarchy produces approximately scale-independent spectral behavior. It does not establish that our observed physics is located at a particular layer, nor that deeper scales are physical lengths. Those conclusions require an operational preparation, a clock and length calibration, an instrument, and empirical data.

Symmetry remains a reservoir of equivalent possibilities, not positive feedback by itself. Feedback requires an instability or gain law. ST248 proves that deterministic equivariance cannot pick one vertex; a genuine spontaneous state source would have to lie outside that class and still explain its outcome probabilities and orientation.

17 Recommended programs ST263–ST277

- ST263.** Test one genuinely new strict symmetry-breaking state source outside deterministic D_{12} -natural rules, or preserve the ST248 no-go.
- ST264.** Classify nonlinear equivariant state-to-multiplication maps by orbit type and stabilizer monotonicity.
- ST265.** Extend only the lowest-clearance branch pair with interval event detection.
- ST266.** Derive a rational dual for an affine qubit channel not unitarily equivalent to amplitude damping or a Pauli mixture.
- ST267.** Build an interval KKT certificate or counterexample for ST252.
- ST268.** Test uniqueness of the flat mean-zero connection under a wider naturality category including refinements.
- ST269.** Seek the first certified nuisance-cover failure beyond halfwidth 0.0005 under a fixed resource cap.
- ST270.** Construct exact finite-confidence regions for ST255 under clustered counts.
- ST271.** Resolve generic misaligned joint Clifford uniqueness or give an explicit two-gap counterexample.
- ST272.** Refine the belief-transfer enclosure with adaptive boxes and optimized Chernoff s .
- ST273.** Add a finite-dimensional controller and account for its cyclic free-energy cost.
- ST274.** Intersect the ST259 cone with a strict refinement functor only if a new one is justified.
- ST275.** Derive finite-cutoff error bounds for the log-Haar plateau and audit trace-class completion.
- ST276.** Design a frozen fiber-resolving instrument specification without claiming laboratory execution.
- ST277.** Resume external validation only with independently registered events, calibration, and custody.

The scientific priority is ST263. The present batch has proved exactly why the localized state cannot be concealed in the symmetric operator. A next attempt is admissible only if it introduces

a genuinely new symmetry-breaking source class. ST275 is the highest-value multiscale follow-up because it tests whether the log-Haar plateau survives a mathematically controlled infinite-refinement completion.

18 Reproducibility

The executable source `fin_st248_st262_research.py` writes the aggregate JSON, CSV ledger, fifteen packets, and eight figures. The replay `test_fin_st248_st262.py` checks packet hashes, live representation dimensions, all 360 continuation inclusions, rational recovery data, the complete 21,952-box count, exact modal covariance, Clifford excess identity, transfer bounds, graph-Laplacian signs, and complement observables. The aggregate JSON records Python 3.12.3, NumPy 1.26.4, SciPy 1.17.1, SymPy 1.12, and random seed 20260812.

ST252 is explicitly numerical. ST248, ST249, ST251, ST253–ST261 are theorem-level only inside their declared mathematical classes. The SHA-256 manifest fixes the release artifacts byte-for-byte.

19 Final conclusion

The current round does not turn FIN into experimentally tested physics. It does replace three vague questions by precise results:

localized preparation is impossible from deterministic symmetric A alone,

nonlinear response has a flat covariant completion on H_0 ,

scale-invariant refinement has log-Haar form and plateau $2\rho \ln 2$.

The remaining boundary is equally precise. A state source must break the ST248 class; a physical refinement must select one point in the ST259 cone and provide a fiber-resolving instrument; a physical scale flow must source the log-Haar density, cutoffs, and units. No such source is presently derived. That separation between achieved mathematics and missing physics is the main scientific outcome of ST248–ST262.